Nicholas Olsen

PhD Candidate, Microbiology, Molecular Biology and Biochemistry

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Research Summary

Doctoral candidate in the Mitchell Lab studying **microglial immunometabolism** in the zebrafish retina - how lipid regulation shapes microglial activation, with focus on lipid droplet biology in retinal microglia and RPE cells and the role of apolipoproteins (e.g. apoc1). Work is relevant to neurodegenerative disease (Alzheimer's, Parkinson's, MS, AMD).

Research

Lipid droplet dynamics in the developing retina (microglia + RPE), via live in vivo spinning-disk confocal imaging with a dedicated LD reporter line.

Apolipoprotein regulation of microglial and retinal lipid handling; apoc1 mutant analysis.

Microglial activation states and metabolic reprogramming (NF-kB transcriptional reporter lines).

Research Experience

Graduate Research Assistant - Mitchell Lab, University of Idaho (2023-present)

Advisor: D. Mitchell. Zebrafish model; live and fixed-tissue imaging (spinning-disk confocal, HCR), ddPCR/qPCR, FACS, microinjection, IHC, cloning. Undergraduate mentorship.

Undergraduate RA / Lab Tech - Crowder Lab, WSU Entomology (2021-2023)

DNA/RNA extraction; brightfield imaging of bumble bee-associated mites (Nikon Eclipse TE300). Co-author, PNW 781 (2024).

Undergraduate RA - Cornejo Lab, WSU Dept. of Biological Sciences (2021)

Population genomics: SNP filtering, species-tree inference (BEAST2, SNAPP, BCFtools, Linux).

Undergraduate Assistant - WSU Bear Research, Education, and Conservation Center (2021)

Grizzly bear husbandry; metabolic assays, blood draws, vitals.

Teaching and Mentoring

Graduate Teaching Assistant - Dept. of Biological Sciences, University of Idaho (2023-present)

BIO 1150 Cells and the Evolution of Life - undergraduate biology lab (foundational concepts).

BIO 3130 Molecular and Cellular Biology - advanced labs for juniors/seniors: ELISA, IHC, RNA/DNA extraction, RT-PCR, gel electrophoresis.

Developed quizzes, exams, and assignments; mentor undergraduate students.

Guest Lecturer - Immunology, University of Idaho (2023-present)

Publications

Olsen N. (manuscript in preparation). Descriptive study of lipid droplet biology in the developing zebrafish retina (retinal microglia and RPE cells). [First author]

Luppino MN, Catron KA, Olsen N, Orpet RJ (2024). Bumble Bee Parasites and Conservation in the Pacific Northwest. PNW 781, Pacific Northwest Extension Publishing. pubs.extension.wsu.edu

Luppino MN, Orpet RJ, Green N, Basu S, Olsen N, Felsot AS (in review). Mite and Pollen Diversity Associated with Wild Bees from Canola and Natural Landscapes in the Palouse Region of Eastern Washington. Journal of Insect Science.

Presentations

Olsen N. Poster, Gordon Research Conference (GRC) on Visual System Development, Newry, ME (2026). COS Travel Grant recipient.

Skills and Techniques

Imaging: Spinning-disk confocal, live in vivo imaging, fixed tissue sectioning, HCR FISH, NIS-Elements, FIJI/ImageJ

Molecular: ddPCR, qPCR, RT-PCR, ELISA, IHC, Gateway cloning, plasmid design, flow cytometry/FACS, zebrafish husbandry and microinjection

Computational: R, Python, Linux/UNIX, BEAST2, SNAPP, BCFtools, TRACER, FigTree

Other: Benchling, Adobe Illustrator

Education

University of Idaho - PhD, Microbiology, Molecular Biology and Biochemistry (2023-present)

Thesis: The Impact of Lipid Regulation in Retinal Microglia Activation and Homeostasis. Committee: D. Mitchell, L. Fortunato, A. Caplan, S. Grieshaber.

Washington State University - BS Biology, Summa Cum Laude (2021)

Wenatchee Valley College - Associates of Science (2018)

Honors

4x Presidential Honor Roll, WSU - Outstanding Senior Award nominee, WSU - College of Science Travel Grant (2026)

Languages

English (native) - Spanish (intermediate listener / novice speaker)

References

Available upon request. [D. Mitchell, UI - D. Crowder, WSU]